



Ahmad Ghalawinji

DATA ENGINEER & AI SPECIALIST

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*Data engineer and AI specialist with industry experience building ML pipelines and data systems at scale.
Seeking data engineering or applied AI roles.*

Experience

Freelance Web & Infrastructure Engineer

Jan. 2025 - Present

SELF-EMPLOYED

Remote, France

- Deployed and managed Linux/WordPress infrastructure for multiple clients with high uptime achieved through proactive monitoring, SSL certificate management, and automated backups.
- Configured Nginx reverse proxies, DNS management, and Git-based deployment workflows to enable rapid iterations and streamlined development processes.
- **Technologies:** Linux, Nginx, MySQL, Git, WordPress, SSL/TLS

Artificial Intelligence Engineering Intern

Feb. 2024 - Aug. 2024

AMADEUS

Nice, France

- Improved production forecasting system by designing data pipeline in Databricks to integrate external pricing signals into the machine learning model.
- Achieved 4% improvement in forecast accuracy, resulting in 0.4% revenue increase.
- **Technologies:** Python, MongoDB, Azure, Spark on Databricks

Machine Learning Intern

Feb. 2023 - Jul. 2023

INRIA

Grenoble, France

- Conducted research into deep neural models including transformers and variational autoencoders (VAEs) to enhance representation learning.
- Utilized PyTorch to develop VAE that reduced data dimensionality, enabling study of dynamic system behavior.
- **Technologies:** Python, PyTorch, VAEs

Software Engineering Intern

Feb. 2022 - Jul. 2022

LABORATOIRE D'INFORMATIQUE DE GRENOBLE (LIG)

Grenoble, France

- Defined automated planning workflows and implemented transition from Domain-Specific Language (DSL) to automated planning using meta-model for Petri nets.
- **Technologies:** DSL, Modelling

Education

MSc in Data Science and Artificial Intelligence (MoSIG M2)

Sep. 2023 - Sep. 2024

ENSIMAG AND GRENOBLE ALPES UNIVERSITY

Grenoble, France

- Relevant coursework: Advanced Machine Learning, Natural Language Processing, Big Data Management

MSc in Informatics (MoSIG M1)

Sep. 2022 - Sep. 2023

ENSIMAG AND GRENOBLE ALPES UNIVERSITY

Grenoble, France

- Relevant coursework: Algorithms, Software Engineering, Operating Systems

BSc in Computer Engineering

Sep. 2016 - Sep. 2021

TISHREEN UNIVERSITY

Latakia, Syria

- Relevant coursework: Data Structures, Computer Architecture, Database Systems

Skills

Programming

Python (Advanced), SQL, Java

Machine Learning & NLP

Supervised & Unsupervised Learning, NLP, RAG, ChromaDB, VAEs, Transformers, PyTorch, Scikit-learn, XGBoost

Data Engineering

Apache Spark, Apache Airflow, Azure Databricks, AWS (S3, EC2), ETL Pipelines, PostgreSQL, MongoDB, Snowflake, DuckDB, Terraform

Visualization

Streamlit, Plotly, Gradio

Tools & Deployment

Git, Linux, Docker, LaTeX

Projects

LegalLens — AI Contract Risk Analyzer | GitHub

2026

- Engineered an async document-processing pipeline (Celery + Redis) ingesting PDF/DOCX contracts, parsing and structuring clauses, then routing each through a LangGraph multi-agent workflow for risk classification and hybrid RAG retrieval (ChromaDB + BM25).
- Exposed the pipeline via a FastAPI REST API with task-polling endpoints and a Next.js 14 frontend; supports pluggable LLM backends (Ollama, Groq, Gemini) with zero paid-API dependency.
- **Technologies:** Python, FastAPI, Celery, Redis, ChromaDB, LangChain, LangGraph, Next.js 14, Docker

AdaptiveRAG-MA — Self-Improving Multi-Agent LLM System | GitHub | Report

2026

- Designed a four-agent orchestration system (Memory, Planner, Executor, Critic) backed by a FAISS vector store for Failure-Driven Experience Replay — task outcomes persisted and retrieved semantically to improve future runs without model retraining.
- Built a FastAPI backend with SSE streaming for real-time agent-step events and structured LoRA training-data export from stored task history.
- **Technologies:** Python, FAISS, FastAPI, Ollama, Docker

Trend Oracle — News Q&A System with RAG | GitHub | Report

2026

- Built a fully local data pipeline collecting articles from five RSS feeds, enriching records with NLP (sentiment, NER), persisting structured data in DuckDB, and indexing embeddings into ChromaDB for RAG-powered natural-language queries with cited sources.
- Containerised the full stack with Docker Compose and delivered a Gradio interface with streaming responses.
- **Technologies:** Python, spaCy, HuggingFace Transformers, ChromaDB, DuckDB, Ollama, Gradio, Docker

Climate Change & Youth Cardiorespiratory Fitness | GitHub | Report

2026

- Ingested and joined ERA5 climate reanalysis data with socioeconomic indicators across 28 European countries; engineered features and trained XGBoost models predicting 4.7% CRF decline at +3°C with 55% recovery potential under policy interventions.
- Built an interactive Streamlit dashboard enabling policymakers to explore climate scenarios and model outputs in real-time.
- **Technologies:** Python, Pandas, Scikit-learn, XGBoost, Plotly, Streamlit, ERA5 Climate Data

Port Calls Detection & Vessel Journey Modelling | GitHub | Report

2025

- Built an ETL pipeline detecting port call events from 27,000+ AIS records across 16 ports using geodesic geofencing and 30-min deduplication; orchestrated via Airflow DAG into PostgreSQL with LEAD() window functions to infer next destinations.
- Delivered a Streamlit dashboard with multi-dimensional filters; containerised the full stack with Docker Compose.
- **Technologies:** Python, PostgreSQL, Apache Airflow, Streamlit, Docker, Pandas, GeoPy

Product Analytics Platform — Music Streaming Pipeline | GitHub

2025

- Generated 1M+ synthetic streaming events for 50,000 users, ingested into a partitioned S3 data lake, then loaded into Redshift with DIS-TKEY/SORTKEY-optimised star schema (fact + dimension marts).
- Delivered a 6-visual QuickSight dashboard (SPICE) covering DAU KPIs, 7-day rolling averages, device breakdown, and top-10 event rankings.
- **Technologies:** Python, AWS S3, Amazon Redshift, Amazon QuickSight

Certificates

- Aug. 2024 **Azure Spark Databricks Essential Training**, LinkedIn Learning
- Aug. 2024 **ETL in Python and SQL**, LinkedIn Learning
- Aug. 2024 **Advanced NoSQL for Data Science**, LinkedIn Learning
- Aug. 2024 **Learning SnowflakeDB**, LinkedIn Learning
- Aug. 2024 **Infrastructure as Code with Terraform**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations: Infrastructure as Code**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations**, LinkedIn Learning
- Aug. 2024 **Agile Foundations**, LinkedIn Learning

Languages

- Arabic** Native
- English** Advanced (C2)
- French** Intermediate (B2)